

The PRE and CLR inputs of a JK flip-flop set and reset the Q output

- ☐ Only after the clock input goes high.
- ☒ Unconditionally.
- ☐ Only if $J = 0$ and $K = 0$.
- ☐ Only if J and K are not connected to anything.

Question 3

Which of the following is correct for a gated D-type flip-flop?

- ☒ The Q output is either SET or RESET as soon as the D input goes HIGH or LOW
- ☐ Only one of the inputs can be HIGH at a time
- ☐ The output toggles if one of the inputs is held HIGH
- ☐ The output complement follows the input when enabled

↳ ⚠ Moving to the next question prevents changes to this answer.

Question 4

Popular application of flip-flop are _____?

- ☒ Cache/Registers
- ☐ Multiplexers
- ☐ Decoders
- ☐ None

⚠️ Moving to the next question prevents changes to this answer.

Question 5

The clock signals are used in sequential logic circuits to ____?

- ☒ Tell when to write to memory
- ☐ Tell how much time has elapsed since the system was turned on
- ☐ Tell the time of the day
- ☐ Carry parallel data signals

⬅️ ⚠️ Moving to the next question prevents changes to this answer.

Question 6

The flip-flops which have not any invalid states are ____?

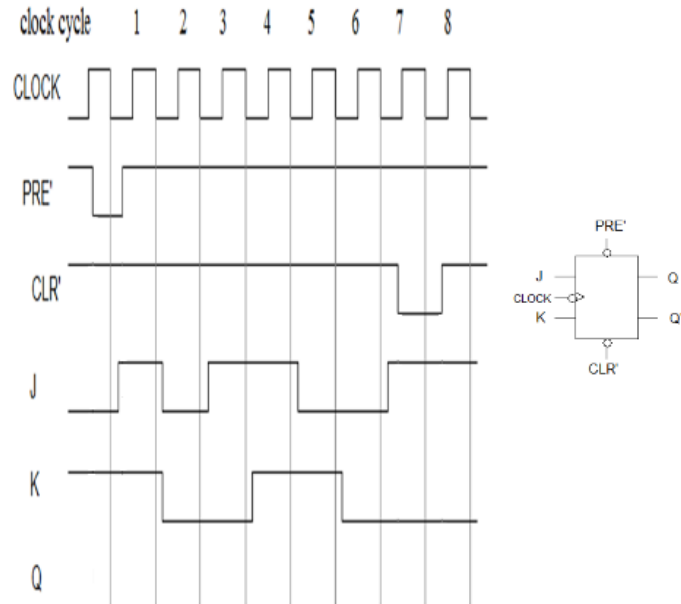
- ☐ S-R, J-K, T
- ☐ S-R, J-K, D
- ☐ J-K, D, S-R
- ☒ J-K, D, T

Question Completion Status:

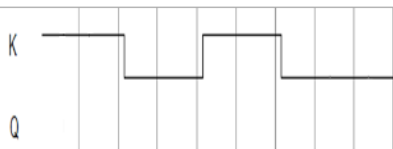
For the JK flip flop and if the values of CLR' and PRE' are as shown in the figure.

Complete the timing diagram. Then answer the question of

What is the value of Q in clock cycles (1, 3, and 5)?



Question Completion Status:



☐ 0 0 0

☒ 1 0 0

☐ None

☐ 1 0 1

Question 8

The main difference between JK and RS flip-flop is that

- ☐ JK flip flop needs a clock pulse
- ☐ JK flip-flop has a Clear-Preset inputs
- ☐ There is a feedback in JK lip-lop
- ☒ JK flip-flop accepts both inputs as 1

Question 9

On a leading edge-triggered S-R flip-flop, the outputs reflect the input condition when

- ☐ The clock pulse transitions from HIGH to LOW
- ☒ The clock pulse transitions from LOW to HIGH
- ☐ The clock pulse is LOW
- ☐ The clock pulse is HIGH

→ ⚠ Click **Submit** to complete this assessment.

Question 10

A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates?

- ☐ AND or NOR gates
- ☐ XOR or XNOR gates
- ☒ NAND or NOR gates
- ☐ None

→ ⚠ Click **Submit** to complete this assessment.

Question 10

The characteristic equation of S-R latch is _____

- ☐ $Q^* = S'R + QR$
- ☐ $Q^* = SR + QR$
- ☒ $Q^* = (S + Q)R'$
- ☐ $Q^* = S'R + Q'R$

→ ⚠ Click **Submit** to complete this assessment.